

COURSEWORK 2 - BRIDGE ACCELERATION ANALYSIS

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1. Dataset & Measurement Selection

The Benchmark Dataset represents a vehicle travelling along a 15 m approach road, a 15 m bridge and a 15 m exit road (45 m total). The vehicle speed is 5 m/s, so it enters the bridge at $t = 3.0$ s and leaves at $t = 6.0$ s. The acceleration records are sampled every 0.001 s, giving a sampling frequency of 1000 Hz. Two bridge measurement locations were selected: $X1 = 7$ m and $X2 = 9$ m. They lie close to the central region of the span and on opposite sides of the 8 m midspan region, providing a useful comparison of spatial variation in bridge vibration while avoiding the low-response support regions.

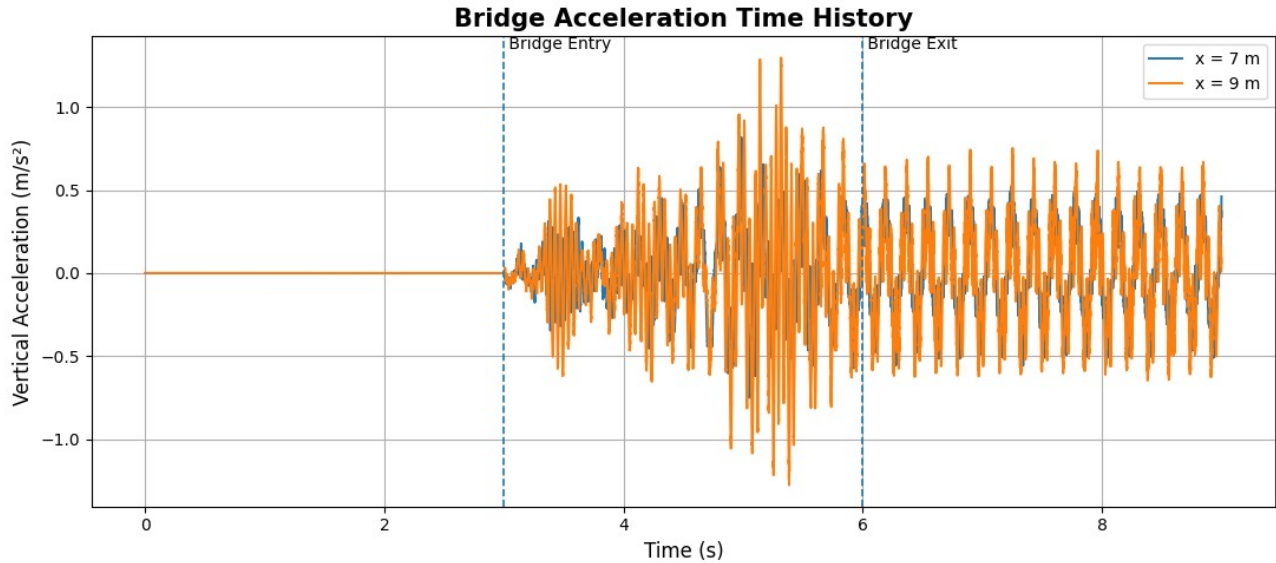


Figure 1. Full acceleration time history at $X1 = 7$ m and $X2 = 9$ m, with bridge entry and exit indicated.

2. Time-Domain Analysis

Figure 1 shows that acceleration is essentially zero before the vehicle enters the bridge. After 3 s, vibration develops at both locations and grows as the vehicle moves across the span. The strongest oscillations occur during the later part of the crossing, around 5.0-5.5 s. Oscillation also remains visible after 6 s, showing that the bridge continues vibrating briefly after the moving load has left.

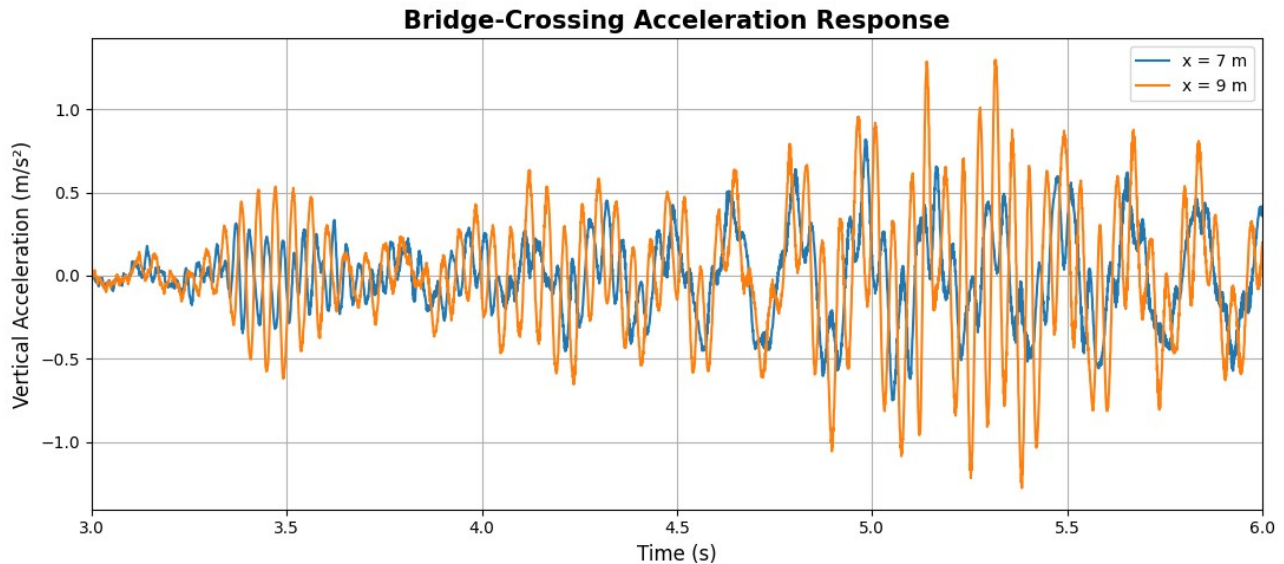


Figure 2. Bridge-crossing acceleration response during the 3-6 s vehicle passage.

During 3-6 s, $X2 = 9$ m shows the stronger response. Its peak absolute acceleration is 1.296592 m/s^2 , compared with 0.818131 m/s^2 at $X1 = 7$ m (about 58.5% higher). The RMS acceleration is also higher at 9 m: 0.381683 m/s^2 versus 0.247130 m/s^2 (about 54.4% higher). These values confirm that the 9 m location experiences greater overall vibration during the vehicle passage.

3. Frequency-Domain Analysis (FFT)

The FFT was calculated using only the 3-6 s bridge-crossing segment. Before the FFT, the signal mean was removed and a Hann window was applied to reduce spectral leakage. Figure 3 compares the acceleration spectra from 0 to 50 Hz.

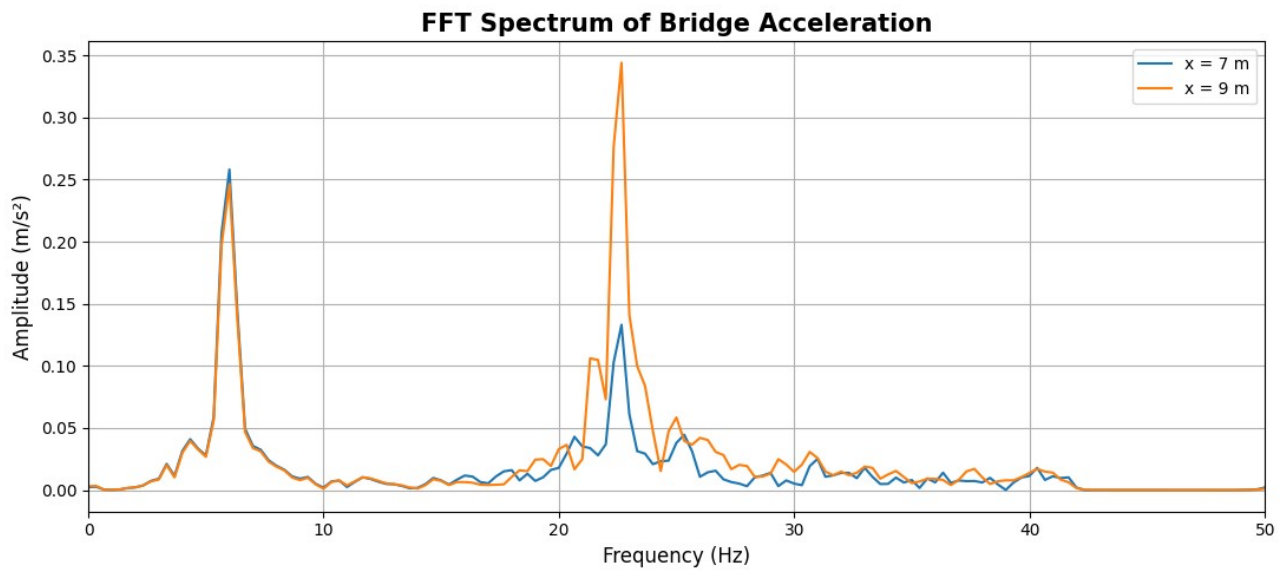


Figure 3. FFT spectrum of bridge acceleration for $X1 = 7$ m and $X2 = 9$ m.

Quantity	$X1 = 7$ m	$X2 = 9$ m
Peak absolute acceleration	0.818131 m/s ²	1.296592 m/s ²
RMS acceleration	0.247130 m/s ²	0.381683 m/s ²
Largest FFT peak	5.998 Hz 0.258072 m/s ²	22.659 Hz 0.343943 m/s ²
Second major FFT peak	22.659 Hz 0.133054 m/s ²	5.998 Hz 0.246405 m/s ²

Additional identified FFT peaks were: $X1 = 7$ m: 25.325 Hz (0.044567 m/s²), 20.660 Hz (0.042978 m/s²), and 4.332 Hz (0.041019 m/s²); $X2 = 9$ m: 21.326 Hz (0.106084 m/s²), 24.992 Hz (0.058478 m/s²), and 25.991 Hz (0.042074 m/s²).

4. Engineering Interpretation

The two locations respond differently in both magnitude and frequency content. $X2 = 9$ m has the higher peak and RMS acceleration, while $X1 = 7$ m has a slightly stronger 5.998 Hz component. Both locations contain strong components at approximately 5.998 Hz and 22.659 Hz, but their relative strengths change with position: the 22.659 Hz amplitude at 9 m (0.343943 m/s²) is about 2.58 times its amplitude at 7 m (0.133054 m/s²). This spatial variation is consistent with bridge mode shapes and vehicle-bridge interaction, because different points along a vibrating structure can have different modal amplitudes. The common peaks may represent important dynamic components of the coupled bridge/vehicle response, but they should not automatically be labelled as bridge natural frequencies; moving-load excitation, vehicle dynamics and road-surface effects can also contribute. Overall, the results show that bridge vibration is position-dependent and that time-domain measures (peak and RMS) and frequency-domain measures (FFT peaks) provide complementary information.

Conclusion

The 9 m location exhibits the stronger bridge-crossing vibration, with a peak acceleration of 1.296592 m/s² and RMS of 0.381683 m/s², compared with 0.818131 m/s² and 0.247130 m/s² at 7 m. The two principal shared frequency components are 5.998 Hz and 22.659 Hz, with the 22.659 Hz component dominant at 9 m. The comparison demonstrates clear spatial variation in bridge vibration under the same vehicle passage.